

Data4HD: Collaborative Open Data for a Transparent and Sustainable Heidelberg

Final Project Report

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Abstract

This report details the "Data4HD" project, a collaborative initiative between SRH University and the City of Heidelberg to establish a sustainable and integrated Open Data infrastructure. The project aimed to enhance transparency, accountability, and data-driven decision-making within the city administration. The methodology involved a comprehensive analysis of Heidelberg's existing open datasets, focusing on three key areas: accessible parking, the impact of weather on mobility, and shifts in political discourse. A comparative analysis framework was employed, benchmarking Heidelberg's data practices against leading German cities, including Bonn and Karlsruhe.

Key findings reveal that while Heidelberg possesses high-quality, dynamic occupancy data for its parking garages, it lags in providing a broad, static inventory of on-street parking facilities compared to Bonn. Data quality assessments identified significant gaps and inconsistencies across several datasets, particularly in descriptive fields and location information. The analysis of council data highlighted a need for more machine-readable formats to improve accessibility, as demonstrated by Karlsruhe's API-driven approach.

Based on these findings, the report provides five strategic recommendations: **(1)** Expand the inventory of on-street parking data; **(2)** Enrich datasets with critical attributes like capacity; **(3)** Extend dynamic data collection to on-street environments; **(4)** Implement robust data quality assurance processes; and **(5)** Actively promote data usage to foster innovation and civic engagement. This project lays a foundational roadmap for Heidelberg to leverage open data for a more transparent and sustainable future.

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Introduction

1.1 Project Background and Motivation

The Data4HD project was initiated to collaboratively establish and continuously enhance a sustainable, scalable, and integrated Open Data infrastructure for the City of Heidelberg. In an era where data is a critical asset for urban management, open data initiatives are paramount for fostering smart city development, enabling citizen science, and promoting transparent governance. The global Open Data movement, supported by policies such as Germany's *Open-Data-Gesetz*, encourages municipalities to make their data publicly available. This fosters an ecosystem where residents, businesses, and researchers can develop innovative solutions to urban challenges, from improving traffic flow to monitoring environmental policy effectiveness.

This project, leveraging a dynamic partnership between SRH University and the City of Heidelberg, aimed to provide students with practical, hands-on experience in the burgeoning field of data-driven urbanism while simultaneously delivering tangible value to the city's administration and its citizens.

1.2 Project Goals and Objectives

The overarching goal was to facilitate robust, evidence-based decision-making within the city administration, significantly enhancing transparency, accountability, and public trust. The project's objectives were structured to address this goal from multiple angles:

- **Analysis and Evaluation:** To conduct a deep and comprehensive analysis of existing datasets on Heidelberg's CKAN Open Data platform. This involved performing exploratory data analysis (EDA), assessing data integrity and consistency, and evaluating metadata completeness.
- **Benchmarking and Best Practices:** To explore national Open Data platforms like Open.NRW and review best practices from leading cities, including Freiburg, Hamburg, Bonn, and Karlsruhe, to establish a benchmark for Heidelberg's strategy.
- **Integration and Expansion:** To identify complementary datasets that could enrich Heidelberg's existing data and to collaboratively define requirements for new data collection based on specific administrative and public needs.

- **Monitoring and Decision Support:** To develop technical workflows for data integration and design frameworks for intuitive dashboards. These tools are intended to represent key municipal indicators related to budgeting, project statuses, and service delivery outcomes for both city officials and the public.

1.3 Scope and Focus Areas

To ensure a targeted and impactful analysis, the project concentrated on three user-centric topics, each framed as a central question to be answered through data:

1. **Case Study Topic 1: Accessible Parking Spaces:** “How well is Heidelberg equipped with accessible parking spaces, and are there areas that need improvement?”
2. **Case Study Topic 2: Impact of Weather on Mobility:** “How do weather conditions, such as temperature and precipitation, affect traffic volume and transportation choices in Heidelberg?”
3. **Case Study Topic 3: Shifts in Political Discourse:** “Which agenda items have shaped political discourse in Heidelberg over the past years, and how have priorities shifted over time?”

Project Methodology

2.1 Agile Framework and Sprint Planning

The project was executed using an agile framework, organized into distinct sprints with clearly defined objectives. This iterative approach was essential for a project involving data exploration, as it allowed the team to remain flexible and adapt to unexpected discoveries, such as data quality issues or the availability of new datasets. The sprints were structured as follows:

- **Sprint 1: Discovery and Scoping.** This initial phase focused on data acquisition via APIs, exploratory data analysis (EDA), systematic logging of data quality issues, and the development of user personas to guide the project.
- **Sprint 2: Modeling and Solution Design.** This phase concentrated on deeper analysis, including the creation of an Entity-Relationship Diagram (ERD) to model data structures, the formulation of a data enrichment strategy to address missing values, and the execution of the formal comparative analysis against other cities.

2.2 User-Centric Design: Developing Personas

To ensure the analysis remained grounded in real-world needs, the team developed two user personas representing key stakeholder groups. These personas were a constant reference point throughout the project, ensuring the final recommendations were practical and valuable.

- **Dr. Evelyn Reed (The Data Detective):** A data scientist representing the academic and research community. Her primary frustration is with poorly documented, inconsistent, and non-machine-readable data, which hinders her ability to conduct scalable and reproducible research. Her needs heavily influenced the project's focus on data quality, API accessibility, and robust metadata.
- **David Miller (The Everyday Resident):** A citizen who represents the general public. He is not a data expert but is interested in how city operations impact his daily life. He wants to use city data for practical tasks like finding a parking spot easily, understanding how his tax money is spent, and staying informed about

policies affecting his neighborhood. His perspective guided the focus on user-friendly applications and transparent data visualization.



Figure 2.1: *User Personas: Dr. Evelyn Reed and David Miller.*

2.3 Comparative Analysis Framework

A core component of the methodology was a comparative analysis framework, systematically benchmarking Heidelberg's open data against other German cities. The evaluation criteria were based on the FAIR data principles (Findability, Accessibility, Interoperability, Reusability) and included:

- **Data Availability:** The scope and variety of datasets offered.
- **Accessibility:** The methods for accessing data (e.g., direct API access vs. manual file downloads).
- **Data Quality:** The completeness, consistency, and accuracy of the data.
- **Documentation and Licensing:** The clarity of metadata, data dictionaries, and data usage licenses.
- **Update Frequency:** How current and regularly maintained the datasets are.

Key findings from this review identified **Bonn's portal** as "**Best Overall**" for its modern interface and strong API, while **Cologne's** was noted as the "**Biggest Weakness**" for being dated and slow.

2.4 Data Acquisition and Processing

Data were acquired programmatically via APIs from the respective city portals where available. The initial data formats were primarily JSON and GeoJSON. To create a

standardized environment for analysis in different sources, all datasets were converted to CSV format using Python scripts. This process also served as the first step in data cleaning and schema normalization.

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Case Study Topic 1: Analysis of Heidelberg's Parking Infrastructure

3.1 Exploratory Data Analysis and Quality Assessment

Initial EDA on Heidelberg's parking datasets revealed several significant quality issues that would prevent reliable, automated analysis.

- **Inconsistent and Ambiguous Data:** The disabled parking dataset was particularly problematic. The 'PARKPLATZ' column, intended for categorization, contained a chaotic mix of German terms ("Behinderte" and the typo "Rehinderte"), time restrictions ("8-17"), and other notes, all within a single unstructured field. This makes programmatic filtering impossible without extensive manual cleaning.
- **Logical and Structural Errors:** The historical datasets for parking garages contained clear data integrity errors. For the "P3 Crown" garage, the 'outOfServiceSlotNumber' field consistently showed a value of **-10**, which is logically impossible. For the "P26 Kirchheim" garage, the 'occupiedSpotNumber' remained at **0** across thousands of timestamps, strongly suggesting a persistent sensor or data pipeline failure.
- **Pervasive Missing Data:** Key datasets were found to be critically incomplete. The main 'parking garage' dataset had nearly **100% of values missing** for 'maximumParkingDuration'. The 'disabled parking' dataset showed major gaps in descriptive fields like 'BESCHREIBU' (description) and 'BESCHRIFTU' (labeling).

id	Notiz	BEZEICHNUNG	BETREIBER	TYP	XTRID	URN	BESCHREIBUNG	BESCHRIFTUNG	PARKPLATZ
01e10d390a735e5b177c7a342de393dc	Ab, mit Basis gefunden	Kirchstraße 20-22	Stadt Heidelberg	öffentlich	45872814_3	F3E			Behinderte
043c829a07b03ebcb4478b0f31b10c9		Mönchgasse / Heiliggeiststraße	Stadt Heidelberg	öffentlich	33077310_1				Behinderte
053a58717092d3d800216e8d8601d6f6		Steubenstr. (Grahampark)/Doss. Landstr.	Stadt Heidelberg	öffentlich	33077229_2	E9A			Behinderte
0586da1bb2270fe43e1425924b718137		Poststraße 17	Stadt Heidelberg	öffentlich	37638360_1				Behinderte
075cf1e07c7c4d94d838c2f29ceaf6		Blumenthalstraße 38	Stadt Heidelberg	öffentlich	33077358_1	F33		Mo-Fr 8-17	Behinderte
08a8109806eade3e02a58646664d83		Untere Neckarstraße (ggü. Nr. 62)	Stadt Heidelberg	öffentlich	37638346_1	F26			Behinderte
09ccc50a0f536c02e778896aaa5d646		Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_1			Mo-Fr 8-17	Behinderte
0b96b3c6e44e29b7715f3d97a785be5a		Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_4			Mo-Fr 8-17	Behinderte
0bbdc0b101f64013b61fc2139745fc479		Hardtstraße 2 (Zentralbetriebshof)	Stadt Heidelberg	öffentlich	33077379_5			beschränkt	Behinderte
0bd248b45ffeff516583cb173d8c08d7		Blumenstraße 38	Stadt Heidelberg	öffentlich	46911969_6			Mo-Fr 8-17	Behinderte
0bec49d871a9a3d7bcb467f5be9ce574c	Baustelle	Rohrbacher Straße (Bergfriedhof)	Stadt Heidelberg	öffentlich	33076962_2			7-19, 2h	Behinderte
0c263c6a23b175d1f65220ac9c8bc10	Schild abgedeckt	In der Neckarhelle 82	Stadt Heidelberg	öffentlich	33077136_1	F0C		Mo-Fr 8-18	Behinderte
0c598587f851364b160967ef6ba5f157	Baustelle	Uferstraße gegenüber Nr. 38	Stadt Heidelberg	öffentlich	33076857_1	E7D			Behinderte
106880d6e6c4d48453da8000f5e1503		Vangerowstraße 4 (Thermalbad)	Stadt Heidelberg	öffentlich	33076896_2	E7B			Behinderte
11ba4dd3fa12cbd1d79d1e42d31985c8	Fläche uneben	Poststraße 5	Stadt Heidelberg	öffentlich	37638363_1				Behinderte
1225ae57672d3d51b3134746d85783ce		Kriegsstraße 16	Stadt Heidelberg	öffentlich	33077316_1			8-19, 2h	Behinderte
12bf42d0f071badb2666b5b1dc080073	Keine Klare Markierung	Tiergartenstraße (Sportzentrum Nord)	Stadt Heidelberg	öffentlich	33076986_1				Behinderte
1300d6ad34b0b1eb752a06158a545733		Dossenheimer Landstraße 18a	Stadt Heidelberg	öffentlich	37638369_1	F30		beschränkt	Behinderte
1424c3892cf5a7adb2b01b01cbf6ea8b		Berheimer Straße 63	Stadt Heidelberg	öffentlich	45872823_1	F31		Mo-Fr 8-19	Behinderte
14e59663f615ede93361b26604822f09	Baustelle	Akademiestraße	Stadt Heidelberg	öffentlich	33076929_3				Behinderte
14e9d305536da7c70ce947aa099e9c19	Weg	Pleikartsforster Straße (Sportz. Süd)	Stadt Heidelberg	öffentlich	33076959_2				Behinderte
153dad0ac58f0ab11d81672d3088b5b0	Fläche uneben	Bahnhof Kirchheim / Hardtstraße	Stadt Heidelberg	öffentlich	45872817_2				Behinderte
17ed0c6825cb14d0a334d295a9a70958		Friedhof Wieblingen	Stadt Heidelberg	öffentlich	33077010_1	F28		7-19, 2h	Behinderte
1935904e1fc1c267a5c688126ce10de2		Bahnhofstraße 36	Stadt Heidelberg	öffentlich	45872820_1	F22			Behinderte
193906de382f3f390d828fb9cb0722f7	Keine Klare Markierung	Tiergartenstraße (Tiergartenschwimmbad)	Stadt Heidelberg	öffentlich	33077475_5				Behinderte
1a77a4df7afd17c666ca4c0ed8fb69c1	Boden uneben	Hauptbahnhof (Willy-Brandt-Platz)	Stadt Heidelberg	öffentlich	33077007_5				Behinderte
1c7994d4c8d260ca208274626ebc0450		Hofmeisterweg (Medizinische Klinik)	UniversitätsKlinikum	öffentlich	37638373_6				Behinderte
1debcb7fdb2f187c97f162919885c5ec	Zu wenig/Keine Klare Mark.	Dantestraße 1	Stadt Heidelberg	öffentlich	33077040_1	F44		Mo-Fr 7-20	Behinderte
1ed94a53094923b876fed5cf19255cf	Zu wenig/Keine Klare Mark.	Hofmeisterweg (Medizinische Klinik)	UniversitätsKlinikum	öffentlich	37638373_7				Behinderte
1ef1f38068fbc481586f98f3c616b0a	Boden uneben	Hauptbahnhof (Willy-Brandt-Platz)	Stadt Heidelberg	öffentlich	33077007_4				Behinderte
2134bb78881457a09c20d590b38bf96		Blunischlistraße 14	Stadt Heidelberg	öffentlich	33077415_1			Mo-Fr 8-12	Behinderte
21d48239ceb1f6a2483c3de2edb5489	Boden uneben	Rohrbacher Straße 81	Stadt Heidelberg	öffentlich	33076905_1			beschränkt	Behinderte
2205f9d0971f0cc0911d925415e21a8f	Gibt es nicht?/Baustelle	Heidelberger Straße 14	Stadt Heidelberg	öffentlich	45872835_1	F13		wtg8-19,2h	Behinderte
77d507ff3b45b1e5394d4bdf015034dd		Im Finkenauernd/Mönchertplatz 33	Stadt Heidelberg	öffentlich	42714752_2				Behinderte

Figure 3.1: Example of Data Inconsistencies in the Disabled Parking Dataset.

3.2 Data Modeling: An Entity-Relationship Approach

To bring structure to disparate datasets and devise a strategy for resolving data gaps, an Entity-Relationship Diagram (ERD) was created. The ERD mapped the relationships between four key entities: ‘Parking Garage Metadata’, ‘OffStreetParking Current Data’, ‘OffStreetParking Historical Data’, and ‘Disabled Parking Spots’. It established that fields such as ‘parkingSiteld’ and ‘name’ could serve as composite keys to link data sets in a one-to-many relationship.

This model serves as a critical blueprint for a unified parking data model for Heidelberg. Its immediate practical application was the formulation of a **Data Enrichment Strategy**: by joining the ‘Parking Garage Metadata’ table with the ‘OffStreetParking Historical Data’ table, it became possible to programmatically fill missing latitude and longitude coordinates in the main metadata table, thus fixing a critical data quality issue.

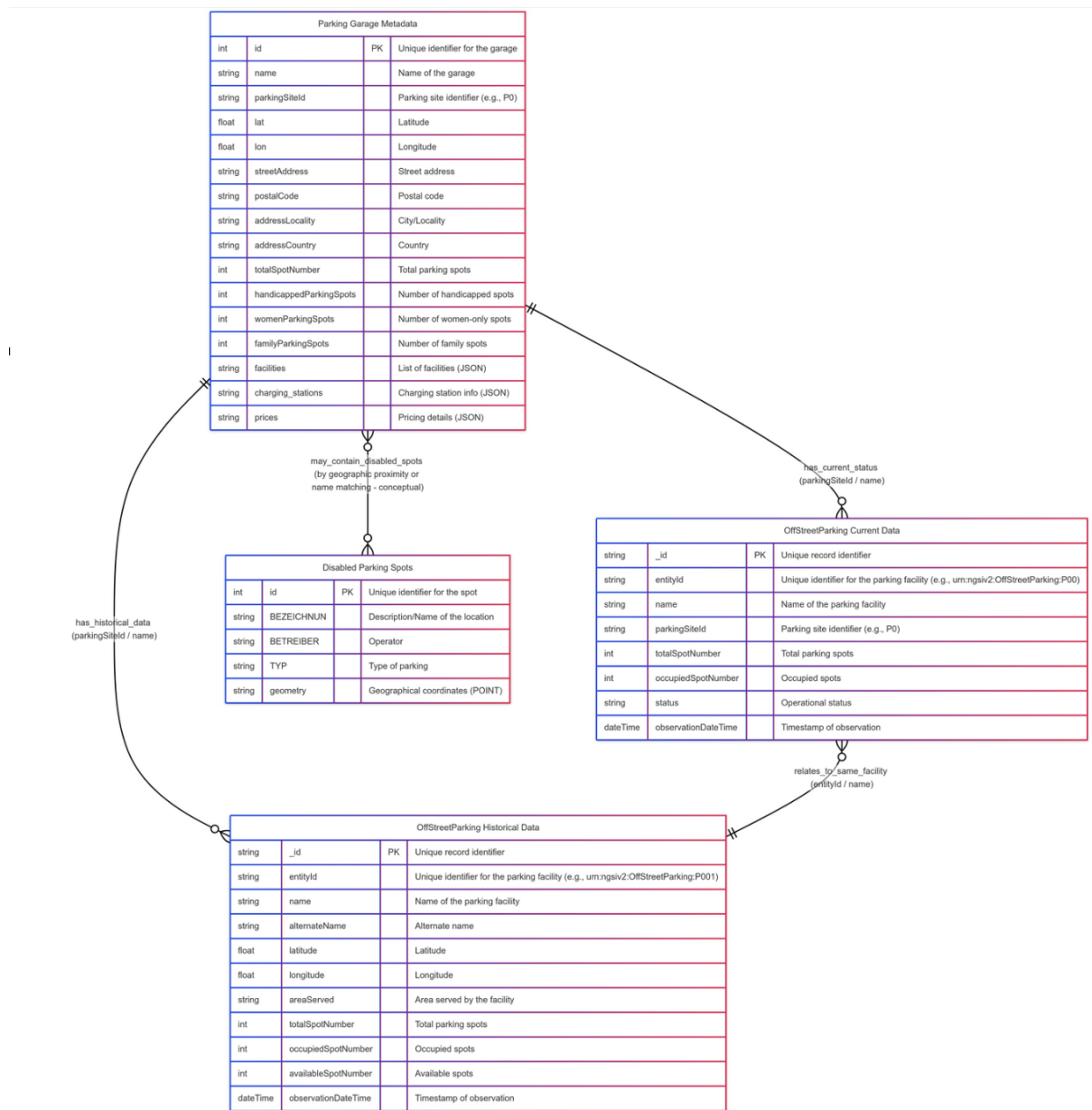


Figure 3.2: Entity-Relationship Diagram for Heidelberg Parking Datasets.

3.3 Comparative Analysis: Heidelberg vs. Bonn

The in-depth comparison with Bonn's data highlighted fundamental differences in open data strategy and maturity.

- **Infrastructure Overview:** Heidelberg provides open data primarily on its **20 garages** and **214 disabled spots**. Bonn, on the contrary, offers a much more extensive inventory, covering **27 garages**, **9 Park & Ride locations**, and **6,632 resident parking zones**. This shows Heidelberg's strength is in dynamic off-street data, while Bonn excels in providing a broad, static inventory of all parking types.
- **Data Assets and Attributes:** Heidelberg's main assets are its rich dynamic and

historical datasets for garages. However, Bonn provides a wider array of static datasets with much richer attributes. For example, Bonn's 'parking bonn koeln osm' dataset contains dozens of useful fields like 'capacity:disabled', 'fee', 'opening hours', 'surface', and 'payment' types, information largely absent from Heidelberg's static data.

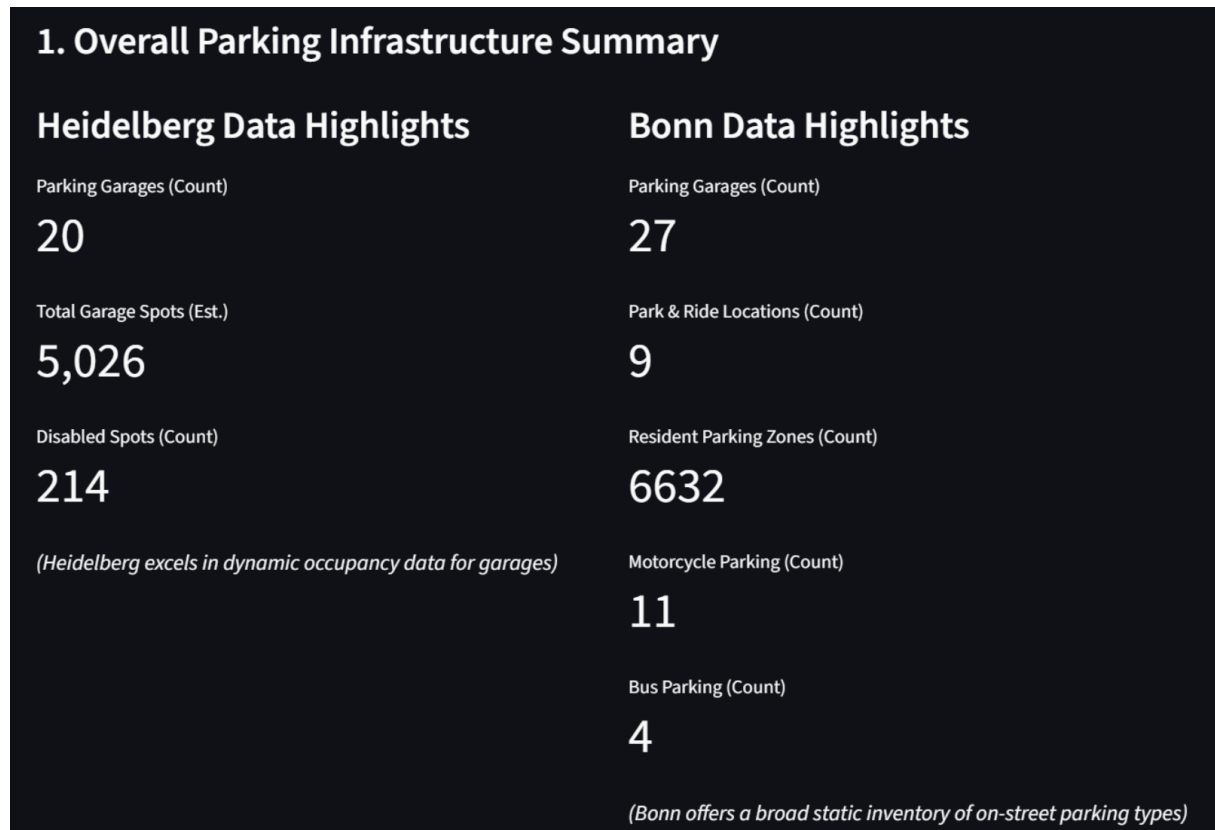


Figure 3.3: *Comparison of Parking Facility Counts: Heidelberg vs. Bonn.*

Case Study Topic 2: Impact of Weather on Mobility & Climate Goals

4.1 Heidelberg's Transportation Mode Share

An analysis of mobility behavior reveals that private cars remain the dominant mode of transportation, accounting for **35%** of trips. However, sustainable modes are significant, with **cycling at 25%**, **public transport at 20%**, and **walking in 15%**. These data are a critical baseline for policymakers to measure the effectiveness of initiatives aimed at promoting sustainable transport, such as the expansion of bike lanes, and to track progress towards the city's goal of a 30% bike share by 2030.

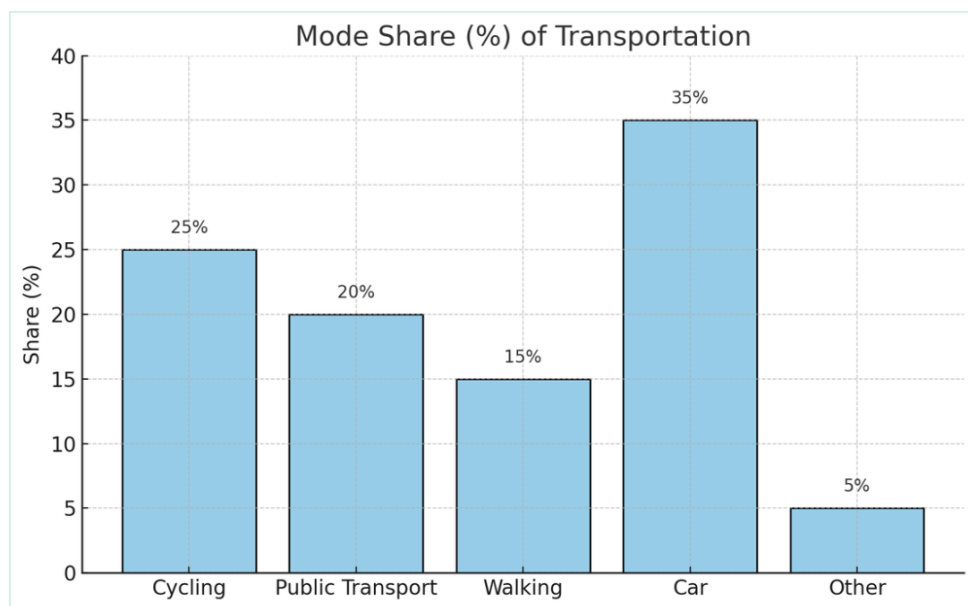


Figure 4.1: *Mode Share (%) of Transportation in Heidelberg.*

4.2 Tracking Climate Goals: CO Emissions and Weather Trends

Heidelberg has demonstrated a strong commitment to climate protection, achieving a **40% reduction in CO emissions since 1990**. This underscores the positive impact of past policies. However, weather data reveals the pressing reality of climate change. The city

has experienced a temperature increase of **+1.2°C over the last 10 years**, and there is a clear trend of increasing annual rainfall between 2015 and 2023. Annual rainfall has risen from 680 mm in 2015 to 740 mm in 2023.

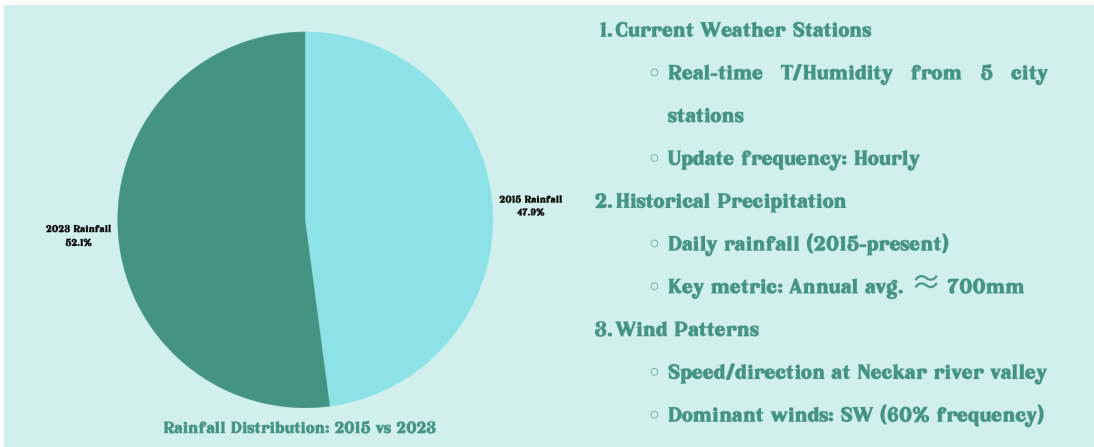


Figure 4.2: Heidelberg Annual Rainfall Trend (2015-2023).

4.3 Data Integration Challenges and Policy Implications

A key challenge was the difficulty in performing granular correlation analysis. Without high-frequency, timestamp-aligned data for traffic volume, public transport ridership, and weather events, it is difficult to precisely quantify the impact of a specific rainstorm on traffic congestion. Despite this, the integrated data is vital for strategic policy-making. The rising precipitation trend directly informs flood prevention strategies and drainage upgrades.

Case Study Topic 3: Analysis of Open Council Data

5.1 Assessing Data Accessibility: Heidelberg vs. Karlsruhe

A comparative analysis was conducted between Heidelberg's and Karlsruhe's open council data portals to evaluate their utility for research and civic engagement. A profound difference in data accessibility philosophy was discovered:

- **Heidelberg:** Provides council data primarily as **PDF documents**. While transparent, this format is designed for human consumption and is difficult to process programmatically.
- **Karlsruhe:** Embraces a modern, developer-friendly approach by offering comprehensive access via a **REST API**, providing data in structured, machine-readable formats like JSON, CSV, and XML.

Evaluation Metric	Heidelberg	Karlsruhe	Analysis & Insight
1. Data Existence	"Yes" (11 Datasets)	"Yes" (Superior Implementation)	Both cities are strongly committed to publishing council data, fulfilling the basic promise of transparency.
2. Machine-Readability	"Poor" (Primarily PDF)	"Excellent" (JSON, CSV, XML, RDF)	Key Differentiator. Heidelberg's PDFs are for human consumption. Karlsruhe's structured formats are for machines , enabling immediate analysis, application development, and data integration.
3. API Access	"No" (Static file downloads)	"Yes" (Comprehensive REST API)	Key Differentiator. Karlsruhe's API allows for live, automated systems. Heidelberg's process is manual. This is the single largest gap in maturity.
4. Time Range	Good (2020 - Present)	Excellent (2014 - Present)	Karlsruhe provides a longer historical record, offering better insights into long-term policy trends.
5. Update Frequency	Good ("Live" dataset available)	Excellent (Real-time via API)	Both cities update data promptly. However, Karlsruhe's API-driven updates are inherently faster and more efficient.
6. Detail Level	Good (Minutes, Members, Calendar, Proposals)	Excellent (Minutes, Members, Agenda Items, Detailed Votes, Documents)	Karlsruhe provides more granular data, including individual agenda items and attachments as separate, downloadable objects, allowing for deeper analysis.
7. License	Good (DL-DE-BY-2.0)	Excellent (DL-DE-BY-2.0)	Both use the same clear, permissive open license ("Data licence Germany – attribution – Version 2.0"), which is ideal for reuse.
8. Ease of Discovery	Good (Searchable on platform)	Excellent (Well-organized with dedicated category)	Data is easy to find on both portals. Karlsruhe's portal has a dedicated "Politics" category, making discovery slightly more intuitive.

Figure 5.1: *Detailed Comparison Framework: Heidelberg vs. Karlsruhe).*

5.2 Impact on Civic Technology and Automated Analysis

The key differentiator is Karlsruhe's API-driven approach, which unlocks the full potential of open government data. It enables developers to build civic tech applications and allows researchers to programmatically track legislative trends. For example, with an API, a researcher could build a script to track the frequency of terms like "Klimaschutz" (climate protection) in council agendas over a decade. A startup could create a service that notifies residents when their street is mentioned in a council document. These possibilities are effectively blocked by Heidelberg's PDF-based system, which requires laborious manual work to extract the same information.

Recommendations and Future Work

6.1 Strategic Recommendations for Heidelberg's Open Data Portal

Based on the comprehensive analysis, five interlocking strategic recommendations are proposed:

1. Expand Data Inventory for On-Street Parking

- **Current State:** The portal lacks data for most on-street parking types.
- **Proposed Action:** Systematically create and publish new datasets for resident parking zones, motorcycle parking, and bus parking, following Bonn's comprehensive model.
- **Expected Impact:** This will provide a holistic view of the city's parking infrastructure, helping different users navigate and supporting comprehensive urban planning.

2. Enrich Datasets with Critical Attributes

- **Current State:** Existing static datasets lack crucial details like capacity.
- **Proposed Action:** Mandate the inclusion of essential attributes like 'capacity', 'tariffs', and 'operating hours' in all new and existing parking datasets.
- **Expected Impact:** This moves the data from being merely informational to being truly functional, enabling better allocation of resources and quantitative analysis.

3. Extend Dynamic Data to On-Street Environments

- **Current State:** Dynamic occupancy data is limited to off-street garages.
- **Proposed Action:** Investigate and pilot methods (e.g., sensors) to collect and publish real-time occupancy data for high-demand on-street parking areas.
- **Expected Impact:** This would be a significant innovation, vastly improving the user experience for drivers and enabling dynamic traffic management strategies.

4. Implement Robust Data Quality Assurance

- **Current State:** Data quality is inconsistent, with significant errors and missing values.
- **Proposed Action:** Establish a formal data governance framework, including automated validation scripts and clear metadata standards.
- **Expected Impact:** Guarantees higher reliability and usability of all open data, building trust with data consumers.

5. Promote Data Usage and Foster Innovation

- **Current State:** The value of open data is not being fully realized due to limited promotion.
- **Proposed Action:** Actively engage with the local tech community and universities through hackathons and clear documentation.
- **Expected Impact:** Fosters a vibrant data-driven ecosystem, leading to the creation of innovative citizen services.

Conclusion

7.1 Summary of Project Achievements

The Data4HD project successfully delivered a strategic roadmap for the evolution of Heidelberg's open data platform. It performed a rigorous, multi-faceted analysis of existing datasets, benchmarked the city's practices against national leaders, and identified critical gaps in data availability, quality, and accessibility. Through detailed case studies, the project has produced a set of actionable, evidence-based recommendations designed to guide the city's future open data strategy toward greater transparency, utility, and civic value.

7.2 Key Takeaways and Final Reflections

The project execution yielded several key takeaways and learnings for future data initiatives.

7.2.1 Key Takeaways

- **Data Availability is a Limiting Factor:** Throughout our analysis, a recurring theme was the desire for more comprehensive datasets. The significant number of null values and gaps in historical records constrained the depth of our potential findings.
- **Data Quality and Language Barriers:** The data was often difficult to process due to quality issues. A persistent challenge was the mix of German and English terms, even within supposedly English-language datasets, which required extensive manual cleaning.
- **The Role of Data Providers:** A major difference observed between Heidelberg and cities with higher quality data was the relationship with data providers. We suggest that creating a partnership program for providers in Heidelberg could significantly increase the volume and quality of available data.

7.2.2 Final Reflections

1. **The most important step is careful research and preparation.** We learned that good planning and a thorough understanding of the data landscape are foundational to a successful project.
2. **Crucial project management and teamwork are the best way to get things done.** Our collaborative, agile approach was essential to navigating challenges and achieving our goals.
3. **Comparing our data against others was the best way to understand our own results and find areas for improvement.** The comparative analysis was invaluable for providing context and identifying tangible improvement targets.
4. **The true value of this work lies in building a foundation for future, more advanced data analysis.** We believe this project has laid the groundwork for a more mature and impactful open data ecosystem in Heidelberg.

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